

A3-EC03-06 · FAILURE DOMAINS

Seven distinct failure domains — each defines an observable effect, operational boundary and explicit non-proof.

F-01 ≠ F-02 ≠ F-03 ≠ F-04 ≠ F-05 ≠ F-06 ≠ F-07 · ALL FAILURE DOMAINS ≠ ENGINEERING COMPONENT FAILURE ≠ BUILDING FAILURE						
FAILURE DOMAIN	RESPONSIBILITY LAYER	OBSERVABLE EFFECT	DATA / CONNECTIVITY CONSEQUENCE	WHAT REMAINS OPERATIONAL	WHAT IT DOES NOT PROVE	RECOVERY / NEXT CHECK
F-01 PHYSICAL DEVICE FAILURE	BUILDING / FIELD	Device silent or reports an implausible value.	That device's observations unavailable or potentially invalid; evidence gap.	Last known building state; observations from other devices.	Engineering component failure or building failure.	Root cause / recovery: unresolved. Buffering alone does not prove device recovery.
F-02 LOCAL POWER FAILURE	BUILDING / FIELD	Multiple devices unreachable together; power correlation if present.	Affected devices' observations unavailable; circuit-wide evidence gap.	Devices on unaffected circuits; last known building state.	Failure of every affected component or the building.	Power restoration / restart / retained buffer behavior: unresolved.
F-03 LOCAL EDGE / AGGREGATION FAILURE	BUILDING / FIELD or JOINT if operator-managed	Multiple devices unreachable together, correlated by physical cluster.	All observations behind that aggregation function unavailable; cluster gap.	Direct-connected devices; other aggregation groups; last known state.	Underlying device, component or building failure.	Recovery not designed. Buffered delivery, if any, preserves original event_time.
F-04 ACCESS / RADIO CONNECTIVITY FAILURE	TBD WITH BEELINE	HouseMaster integration state becomes UNREACHABLE or DEGRADED.	Observations temporarily unavailable or delayed; temporary evidence gap.	Last known building state; local observation/buffering where supported.	Root cause, device/component failure or building failure.	Forward buffered observations after recovery; preserve event_time. Root cause / values: TBD.
F-05 OPERATOR TRANSPORT FAILURE	BEELINE	Correlated connectivity loss beyond one building's pattern.	Observations across shared paths unavailable/delayed; broad evidence gap/backlog.	Last known building state; local observation/buffering where supported.	Device/component/building failure or engineering state change.	Forward buffered observations after recovery; preserve event_time. Retry/SLA: undefined.
F-06 HOUSEMASTER INTEGRATION EDGE FAILURE	HOUSEMASTER	Ingress/processing health signals; connectivity may remain healthy.	New observations cannot be ingested; acceptance delayed; ingestion-side gap.	Already-ingested building state; upstream device/connectivity functions.	Beeline transport, device, component or building failure.	Recovery not designed. Delayed buffered delivery preserves original event_time.
F-07 HOUSEMASTER PROCESSING FAILURE	HOUSEMASTER	Processing backlog or latency.	Domain interpretation and workflow progression may be delayed.	Raw ingested observations if durably stored; existing building state.	Device/connectivity, engineering component or building failure.	Processing recovery undefined. Raw data remains only if durably stored.

HOUSEMASTER INTEGRATION STATES — NOT BEELINE INTERNAL NETWORK STATES

UNKNOWN · PROVISIONED · REACHABLE · UNREACHABLE · DEGRADED · SUSPENDED · DECOMMISSIONED

UNREACHABLE = EXPECTED PATH NOT RESPONDING · ROOT CAUSE UNCLASSIFIED

UNREACHABLE is not mapped to one failure domain and does not identify device, power, edge, access, transport or integration root cause.

TIMING / STORE-AND-FORWARD — SOURCE-SUPPORTED

event_time CREATED → LOCAL BUFFER / STORE → TRANSPORT UNAVAILABLE → CONNECTIVITY RESTORED → BUFFERED OBSERVATION FORWARDED → received_time CREATED

event_time ≠ received_time WHEN BUFFERED · DELAYED DELIVERY DOES NOT REWRITE event_time

CONNECTIVITY LOSS MAY CREATE TEMPORARY EVIDENCE GAP · CONNECTIVITY LOSS ≠ ENGINEERING FAILURE

Buffer size: NOT DEFINED · Retention: NOT DEFINED · Retry: NOT DEFINED · Heartbeat / timeout / polling: NOT DEFINED

DEVICE OFFLINE ≠ COMPONENT FAILED

EDGE OFFLINE ≠ BUILDING FAILURE

NETWORK HEALTH ≠ BUILDING HEALTH

CONNECTIVITY STATE ≠ ENGINEERING STATE